

# Division: Group Size or Group Count?

## Answer Key

### Grade 3

#### Quick Answers

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|---------------|-------------------------|
| 1. 4          | 5. 7                    |
| 2. 3          | 6. stickers each = 9, — |
| 3. 5          | 7. (b) = 6, —           |
| 4. (b) = 4, — | 8. (c) = 6, —           |
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#### What is verified

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4 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 4, 6, 7, 8. The worked solution states what a correct response must contain.

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#### Curriculum

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**Level:** Grade 3

3.OA.A.2 – problems 1, 2, 4, 6, 7, 8

3.OA.A.3 – problems 3, 5, 6, 8

*Difficulty 1–4 of 5 across 13 tagged checks.*

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#### Common wrong answers

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1. If they got 3: answered with the number of bags instead of the cookies in each bag
  2. If they got 5: answered with the number of balls in each can instead of the number of cans
  3. If they got 4: answered with the number of plates instead of the grapes on each plate
  4. If they got 6: repeated the number of boxes as the answer
  5. If they got 4: answered with the wheels on each wagon instead of the number of wagons
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- 1.** 12 cookies shared equally into 3 gift bags. How many cookies in each bag?

**Solution:** The 3 is the number of bags (the group count); the question asks for the group size.  $12 \div 3 = 4$ , because  $3 \times 4 = 12$ . Answering 3 would just repeat the number of bags.

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| 4 |
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- 2.** 15 tennis balls, 5 balls in each can. How many cans?

**Solution:** Here the 5 is the group *size*; the question asks for the group *count*.  $15 \div 5 = 3$ , because  $5 \times 3 = 15$ . The coach fills 3 cans — 5 is how many go *in* each can, not the answer.

3

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**3.** 20 grapes equally onto 4 plates. How many grapes on each plate?

**Solution:** The 4 counts plates (groups); the answer must count grapes per plate (group size).  $20 \div 4 = 5$ , because  $4 \times 5 = 20$ .

5

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**4.** Jo says each of the 6 boxes holds 6 crayons (24 crayons in all).

**Solution (a):** Jo repeated a number that was already given — the 6 boxes — instead of dividing. The problem gives the group count; the question asks for the group size, which must come from  $24 \div 6$ . A correct answer says Jo never divided, and that 6 counts boxes, not crayons.

**Solution (b):**  $24 \div 6 = 4$ , because  $6 \times 4 = 24$ . Each box holds 4 crayons.

4

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**5.** 28 wheels, 4 wheels per wagon. How many wagons?

**Solution:** The 4 is the group size (wheels per wagon); the question asks how many groups.  $28 \div 4 = 7$ , because  $4 \times 7 = 28$ . The toy maker builds 7 wagons.

7

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**6.** 18 stickers shared equally between 2 friends: how many each, and what does the answer count?

**Solution:**  $18 \div 2 = 9$ , because  $2 \times 9 = 18$ .

9

The 9 counts stickers *per friend* — the group size. The group count (2 friends) was given in the story, so a correct second answer names “the number of stickers each friend gets.”

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**7.** Ben: “ $30 \div 5 = 6$ , so each pouch holds 6 marbles.”

**Solution (a):** Ben’s arithmetic is right but his *meaning* is wrong. He divided by the group size (5 marbles per pouch), so the quotient 6 counts *pouches*, not marbles per pouch. Each pouch holds the given 5 marbles. A correct answer says the 6 is the group count, not the group size.

**Solution (b):** Ben fills  $30 \div 5 = 6$  pouches.

6

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**8.** Two stories for  $24 \div 4$ , then the quotient.

**Solution (a):** Sharing story — the 4 must be the number of groups, e.g. “24 treats are shared equally among 4 friends. How many treats does each friend get?” The answer, 6, counts treats *in each group*.

**Solution (b):** Grouping story — the 4 must be the size of each group, e.g. “24 treats are packed 4 to a box. How many boxes are filled?” The answer, 6, counts *groups*.

**Solution (c):**  $24 \div 4 = 6$ , because  $4 \times 6 = 24$ .

6